

Lab_10

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Introduction to the RCSB Protein Data Bank (PDB)

PDB statistics

Download CSV File

```
# Loading pdb stats file onto R

pdb_data <- read.csv("pdb_stats.csv")
pdb_data
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	84	32	214078			
2	1	0	13981			
3	0	0	15759			
4	3	1	4941			
5	0	0	237			
6	0	4	22			

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

Answer Q1: As shown below, **93.79%** of the structures in the PDB are solved by X-Ray and Electron Microscopy.

```
# Count the structures solved by X-Ray
xray_total <- sum(pdb_data$X.ray)

# Count the structures solved by Electron Microscopy
em_total <- sum(pdb_data$EM)

# Count the total structures solved for in PDB dataset
total_total <- sum(pdb_data$Total)

# Find percentage by adding X-Ray & EM then diving by total structures
percent_combined <- ((xray_total + em_total) / total_total) * 100
percent_combined
```

```
[1] 93.7892
```

Q2: What proportion of structures in the PDB are protein?

Answer Q2: As shown with the calculations below, **97.91%** of the structures in the PDB are proteins; this includes pure proteins (such as protein_only) and other combination of complexes that have *partial* proteins (such as protein_oligosaccharide, and protein_na).

```
# Assign row names
rownames(pdb_data) <- pdb_data$Molecular.Type

# Create a Vector for the total "Protein_ONLY" structures
protein_only <- pdb_data["Protein (only)", "Total"]

# Create a Vector for the total "Protein/Oligosaccharide" structures
protein_oligosaccharide <- pdb_data["Protein/Oligosaccharide", "Total"]

# Create a Vector for the total "Protien_NA" structures
protein_na <- pdb_data["Protein/NA", "Total"]

#Calculate percentage of combined proteins using vectors created above
percent_protein <- (((protein_only + protein_oligosaccharide + protein_na) / total_total) * 100)
percent_protein
```

```
[1] 97.9118
```

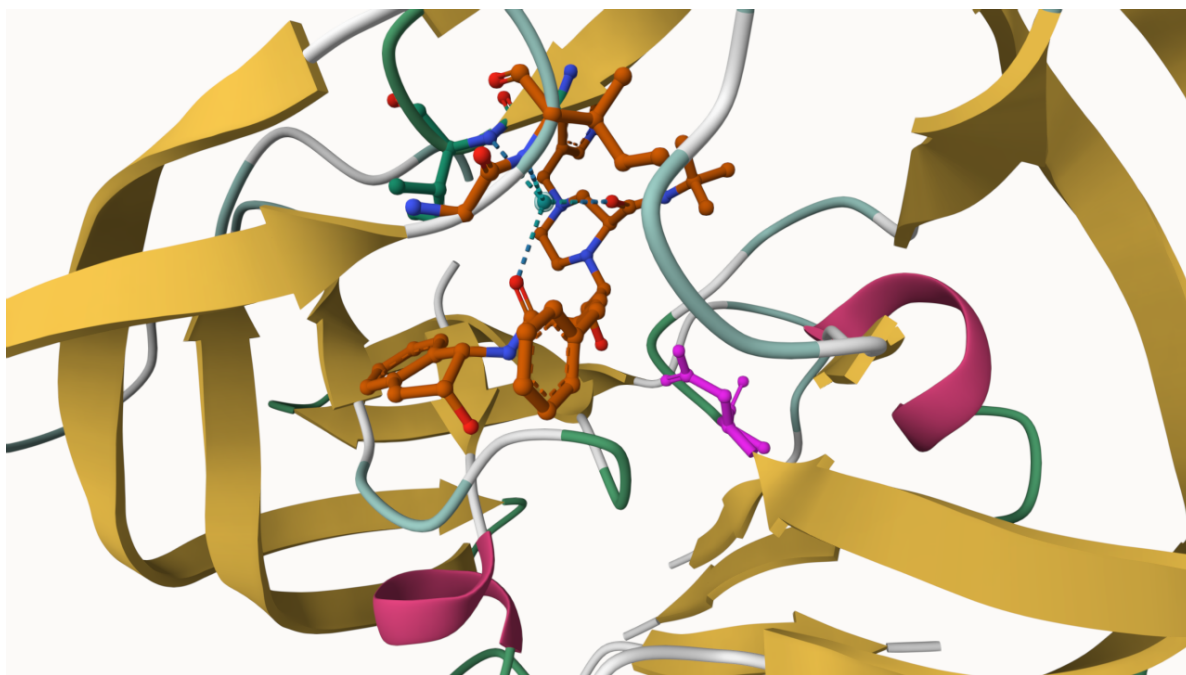
Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Answer Q3: There are **718** HIV-1 protease structures in the current PDB website / database. PDB Results: “This query matches 718 Structures”.

Visualizing the HIV-1 protease structure

Generate and Insert an image of the HIV-cartoon polymer, showing the inhibitor, ASP25, and water 308

```
knitr::include_graphics("1HSG_Ball&Stick_Gradescope.png")
```



The important role of water

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Answer Q4: In the structure provided by Molstar, we only see one atom used to represent each water molecules rather than its 3 atoms (2 Hydrogens and 1 Oxygen). This happens because the EM technique uses electron density to “stipple” the shade of electrons in the molecules configuration. In water, the singular electrons from

each hydrogen are being towards oxygen due to oxygen's higher electronegativity. In other words, while oxygen portrays a dense cloud of atoms the hydrogen atom is not dense hence why the hydrogens are often missed in the stipple configuration of water created by EM.

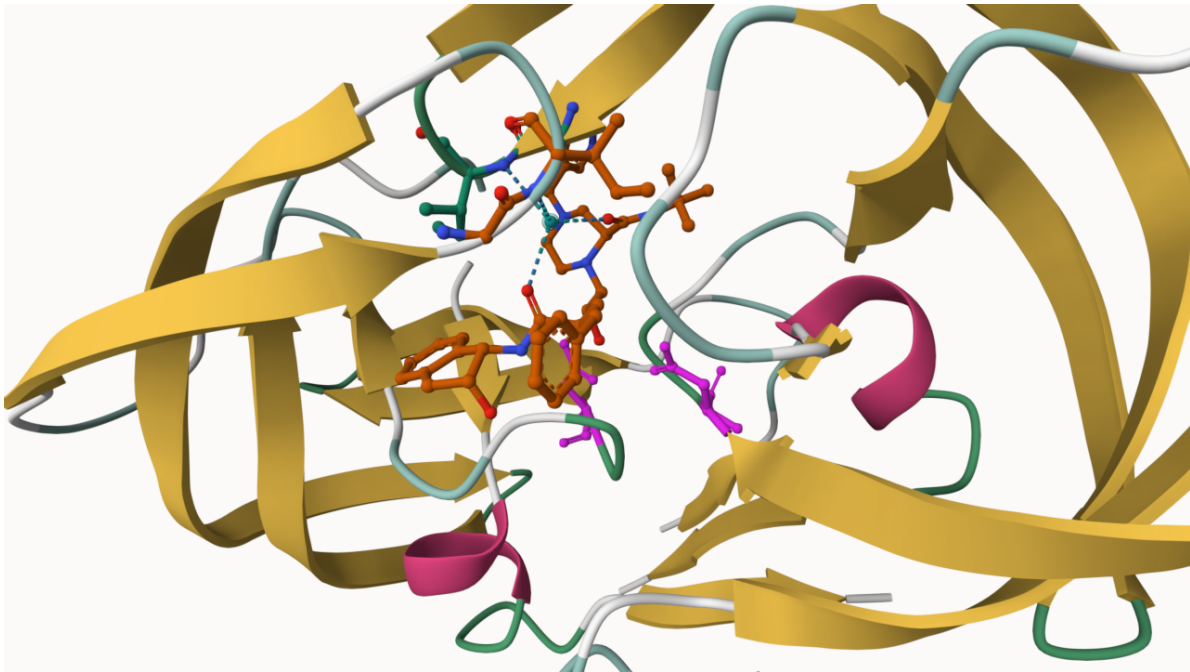
Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

Answer Q5: Yes, there is a critical “conserved” water molecule in the binding site of 1HSG. The this conserved water molecule is found in between 2 oxygens of the 1HSG's ligand. Its residue number is HOH: 308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

Answer Q6: Saved images provided below. The first image shows the two HIV protease chains, long with the ligand(orange), the two catalytic residues ASP25 (magenta, one per chain), and the critical water component(aqua-blue); all of these represented as ball and stick. The second image shows these same components but the ligand is shown with orange spacefill, both Asp25 are shows with pink(magenta) spacefill, and the conserved water is shown with blue spacefill. I prefer the second image as these globular structures are simpler to see and it is easier to spot the conserved water atom and its location in the binding site.

```
# Ball and Stick Image
knitr::include_graphics("1HSG_Ball&Stick.png")
```



```
# Spacefill Image
knitr::include_graphics("1HSG_Spacefill.png")
```



Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is

composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

Answer Q7: SKIPPED

Introduction to Bio3D in R

Since Bio3D was installed at the beginning of class, now its time to just call it out in R.

```
# Call Out Bio3D
library(bio3d)
```

Reading PDB file data into R

```
# Load summary of contents in pdb object
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
  Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

Answer Q7: There are a total of **198 amino acid residues** in this pdb object.

Q8: Name one of the two non-protein residues?

Answer Q8: One of the two non-protein residues is **HOH** (water).

Q9: How many protein chains are in this structure?

Answer Q9: There are **2 protein chains** in this structure.

Attributes:

```
# Find the attributes of the pdb file
```

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

Find atom attribute

```
# Call out the atom attribute
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40
	segid elesy charge												
1	<NA>		N	<NA>									
2	<NA>		C	<NA>									

```

3 <NA>      C   <NA>
4 <NA>      O   <NA>
5 <NA>      C   <NA>
6 <NA>      C   <NA>

```

Predicting functional motions of a single structure

```

# Read PDB structure of Adenylate Kinase
adk <- read.pdb("6s36")

```

Note: Accessing on-line PDB file
 PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```

MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG

```

```

+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call

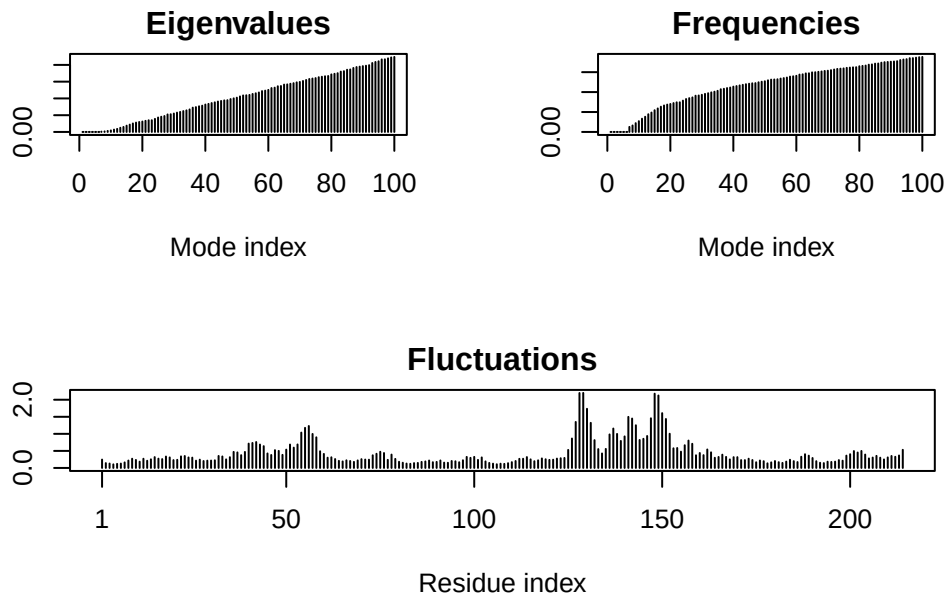
```

Plot the flexibility predictions for adk structure


```
# Perform flexibility prediction
m <- nma(adk)
```

```
Building Hessian...      Done in 0.019 seconds.
Diagonalizing Hessian... Done in 0.464 seconds.
```

```
plot(m)
```



Comparative structure analysis of Adenylate Kinase

Setup

Install packages in the R console NOT your Rmd/Quarto file

Q10: Which of the packages above is found only on BioConductor and not CRAN?

Answer Q10: The **msa** package is found only on BioConductor and not on CRAN.
BiocManager::install("msa")

Q11: Which of the above packages is not found on BioConductor or CRAN?

Answer Q11: The **bio3dview** package was not found on BioConductor or CRAN.
`remotes::install_github("bioboot/bio3dview")`

Q12: True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

Answer Q12: TRUE because the pak package is used as a replacement for older installation methods such as those from GitHub and BitBucket.

Search and retrieve ADK structures

```
# identify related structures to our query Adenylate kinase
library(bio3d)
aa <- get.seq("1ake_A")
```

Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```
      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLVT
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

     121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

fasta

Alignment dimensions:

1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call

Q13: How many amino acids are in this sequence, i.e. how long is this sequence?

Answer Q13: There are a total of **214 amino acids** in this sequence.

Optional:

```
# Blast or hmmer search
#b <- blast.pdb(aa)
# Plot a summary of search results
#hits <- plot(b)
# List out some 'top hits'
#head(hits$ pdb.id)
```

Required:

```
hits <- NULL
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6HPR_A')
```

```
# Download related PDB files
files <- get.pdb(hits$ pdb.id, path="pdbc", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbc", split = TRUE, gzip = TRUE):
pdbc/3HPR.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

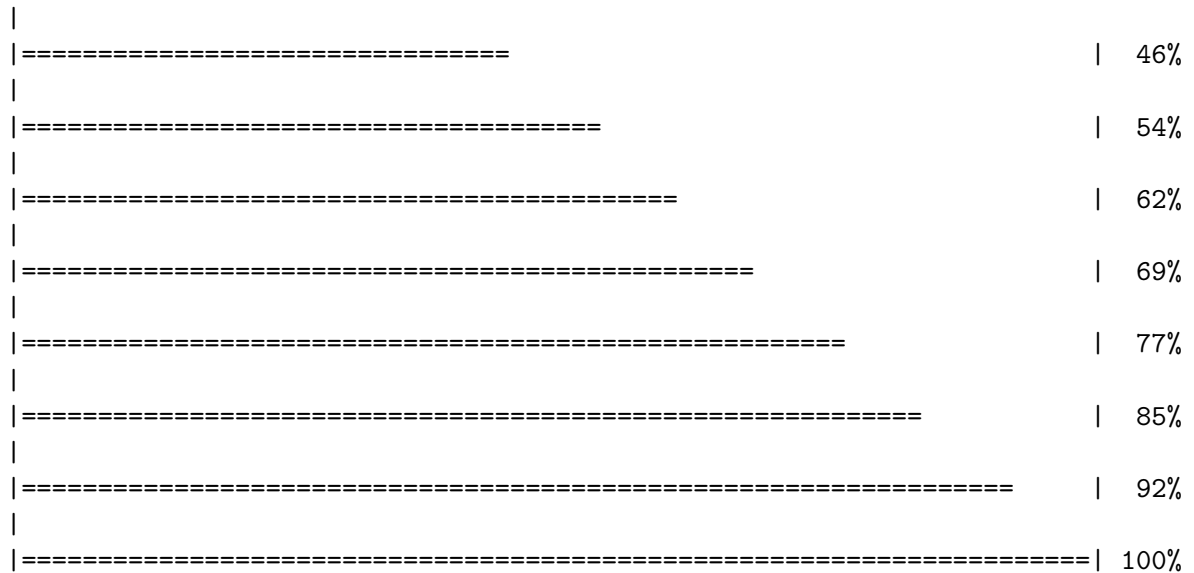
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download

	0%
=====	8%
=====	15%
=====	23%
=====	31%
=====	38%



Align and superpose structures

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
```

```
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
```

```

..   PDB has ALT records, taking A only, rm.alt=TRUE
....  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdbc/split_chain/4PZL_A.pdb

```

Annotate collected PDB structures

```

# Vector containing PDB database codes
ids <- basename.pdb(pdbc$id)

anno <- pdb.annotate(ids)
unique(anno$source)

```

```

[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli 0139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"

```

[5] "Photobacterium profundum"
 [6] "Burkholderia pseudomallei 1710b"
 [7] "Francisella tularensis subsp. tularensis SCHU S4"

View anno
[anno](#)

	structureId	chainId	macromoleculeType	chainLength	experimentalTechnique
1AKE_A	1AKE	A	Protein	214	X-
ray					
6S36_A	6S36	A	Protein	214	X-
ray					
6RZE_A	6RZE	A	Protein	214	X-
ray					
3HPR_A	3HPR	A	Protein	214	X-
ray					
1E4V_A	1E4V	A	Protein	214	X-
ray					
5EJE_A	5EJE	A	Protein	214	X-
ray					
1E4Y_A	1E4Y	A	Protein	214	X-
ray					
3X2S_A	3X2S	A	Protein	214	X-
ray					
6HAP_A	6HAP	A	Protein	214	X-
ray					
6HAM_A	6HAM	A	Protein	214	X-
ray					
4K46_A	4K46	A	Protein	214	X-
ray					
3GMT_A	3GMT	A	Protein	230	X-
ray					
4PZL_A	4PZL	A	Protein	242	X-
ray					
	resolution	scopDomain	pfam		
1AKE_A	2.00	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
6S36_A	1.60	<NA>	Adenylate kinase, active site lid (ADK_lid)		
6RZE_A	1.69	<NA>	Adenylate kinase, active site lid (ADK_lid)		
3HPR_A	2.00	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4V_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		
5EJE_A	1.90	<NA>	Adenylate kinase, active site lid (ADK_lid)		
1E4Y_A	1.85	Adenylate kinase	Adenylate kinase, active site lid (ADK_lid)		

3X2S_A	2.80	<NA>	<NA>
6HAP_A	2.70	<NA>	Adenylate kinase, active site lid (ADK_lid)
6HAM_A	2.55	<NA>	Adenylate kinase, active site lid (ADK_lid)
4K46_A	2.01	<NA>	Adenylate kinase, active site lid (ADK_lid)
3GMT_A	2.10	<NA>	<NA>
4PZL_A	2.10	<NA>	Adenylate kinase (ADK)

	ligandId
1AKE_A	AP5
6S36_A	CL (3),NA,MG (2)
6RZE_A	NA (3),CL (2)
3HPR_A	AP5
1E4V_A	AP5
5EJE_A	AP5,CO
1E4Y_A	AP5
3X2S_A	JPY (2),AP5,MG
6HAP_A	AP5
6HAM_A	AP5
4K46_A	ADP,AMP,PO4
3GMT_A	SO4 (2)
4PZL_A	CA,FMT,GOL

	ligandName
1AKE_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6S36_A	CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A	SODIUM ION (3),CHLORIDE ION (2)
3HPR_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
1E4V_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
5EJE_A	BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
3X2S_A	N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
6HAM_A	BIS(ADENOSINE)-5'-
PENTAPHOSPHATE	
4K46_A	ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A	SULFATE ION (2)
4PZL_A	CALCIUM ION,FORMIC ACID,GLYCEROL

	source
1AKE_A	Escherichia coli
6S36_A	Escherichia coli

6RZE_A	Escherichia coli
3HPR_A	Escherichia coli K-12
1E4V_A	Escherichia coli
5EJE_A	Escherichia coli 0139:H28 str. E24377A
1E4Y_A	Escherichia coli
3X2S_A	Escherichia coli str. K-12 substr. MDS42
6HAP_A	Escherichia coli 0139:H28 str. E24377A
6HAM_A	Escherichia coli K-12
4K46_A	Photobacterium profundum
3GMT_A	Burkholderia pseudomallei 1710b
4PZL_A	Francisella tularensis subsp. tularensis SCHU S4

1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIB

6S36_A

6RZE_A

3HPR_A

1E4V_A

loop

5EJE_A

1E4Y_A

loop

3X2S_A

conjugated adenylate kinase

6HAP_A

6HAM_A

4K46_A

3GMT_A

4PZL_A

Cryst

The crys

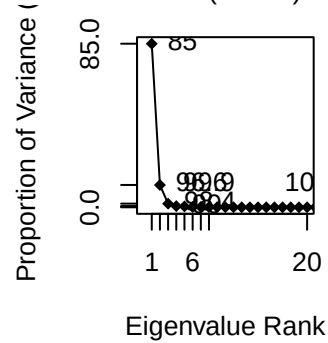
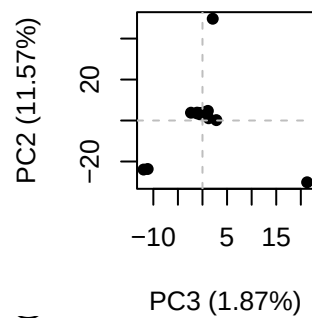
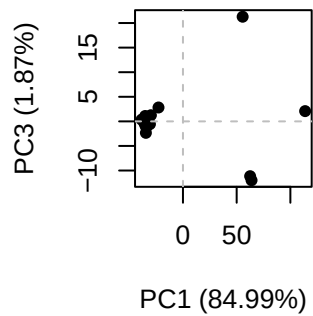
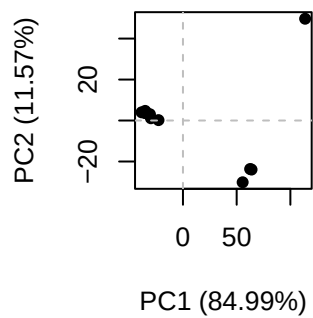
		citation	rObserved	rFree
1AKE_A	Muller, C.W., et al. J Mol Biology (1992)	0.19600	NA	
6S36_A	Rogne, P., et al. Biochemistry (2019)	0.16320	0.23560	
6RZE_A	Rogne, P., et al. Biochemistry (2019)	0.18650	0.23500	
3HPR_A	Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)	0.21000	0.24320	
1E4V_A	Muller, C.W., et al. Proteins (1993)	0.19600	NA	
5EJE_A	Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)	0.18890	0.23580	
1E4Y_A	Muller, C.W., et al. Proteins (1993)	0.17800	NA	
3X2S_A	Fujii, A., et al. Bioconjug Chem (2015)	0.20700	0.25600	
6HAP_A	Kantaev, R., et al. J Phys Chem B (2018)	0.22630	0.27760	
6HAM_A	Kantaev, R., et al. J Phys Chem B (2018)	0.20511	0.24325	
4K46_A	Cho, Y.-J., et al. To be published	0.17000	0.22290	
3GMT_A	Buchko, G.W., et al. Biochem Biophys Res Commun (2010)	0.23800	0.29500	
4PZL_A	Tan, K., et al. To be published	0.19360	0.23680	

rWork spaceGroup

1AKE_A	0.19600	P	21	2	21
6S36_A	0.15940	C	1	2	1
6RZE_A	0.18190	C	1	2	1
3HPR_A	0.20620	P	21	21	2
1E4V_A	0.19600	P	21	2	21
5EJE_A	0.18630	P	21	2	21
1E4Y_A	0.17800	P	1	21	1
3X2S_A	0.20700	P	21	21	21
6HAP_A	0.22370	I	2	2	2
6HAM_A	0.20311				
		P	43		
4K46_A	0.16730	P	21	21	21
3GMT_A	0.23500	P	1	21	1
4PZL_A	0.19130				
		P	32		

Principal component analysis

```
# Perform PCA
pc.xray <- pca(pdbbs)
plot(pc.xray)
```

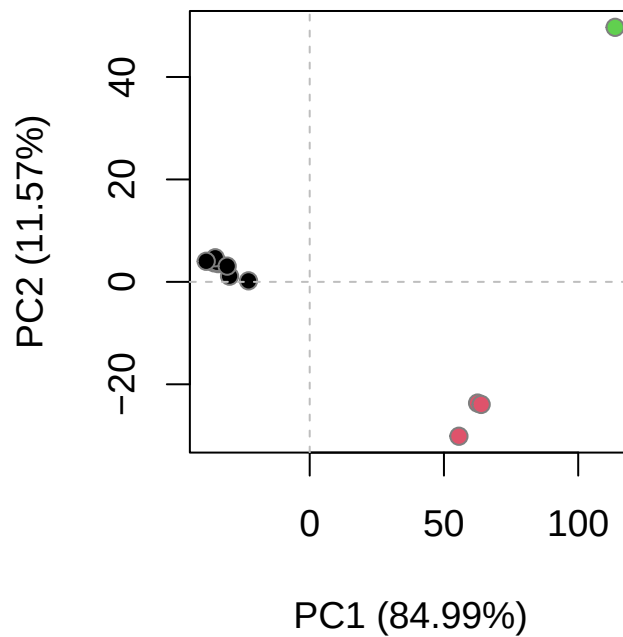


```
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



PCA visualization

```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

Normal mode analysis [optional]

```
# NMA of all structures
modes <- nma(pdfs)
```

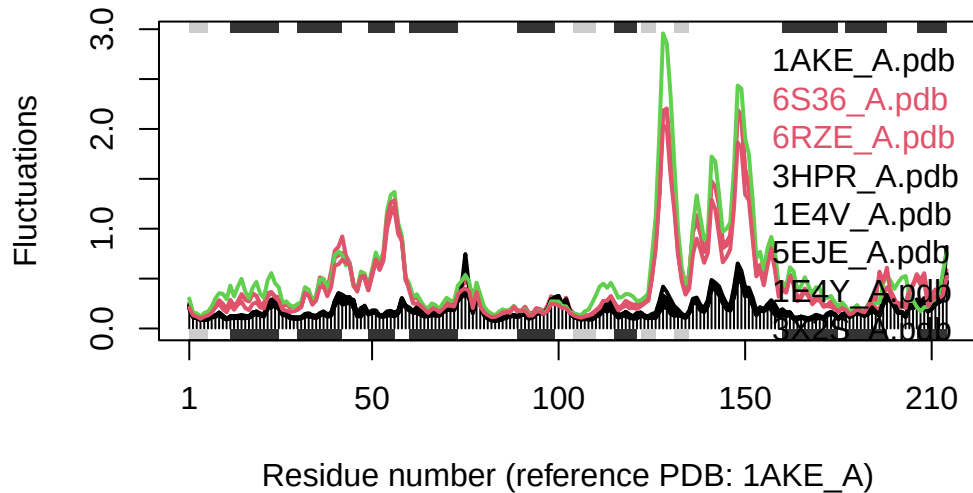
Details of Scheduled Calculation:

```
... 13 input structures
... storing 606 eigenvectors for each structure
... dimension of x$U.subspace: ( 612x606x13 )
... coordinate superposition prior to NM calculation
... aligned eigenvectors (gap containing positions removed)
... estimated memory usage of final 'eNMA' object: 36.9 Mb
```

	0%
=====	8%
=====	15%
=====	23%
=====	31%
=====	38%
=====	46%
=====	54%
=====	62%
=====	69%
=====	77%
=====	85%
=====	92%
=====	100%

```
# Plot
plot(modes, pdb, col=grps.rd)
```

Extracting SSE from pdb's sse attribute



Q14: What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

Answer Q14: This plot displays the relationship between the residue number (its position) and its corresponding fluctuation (degree of flexibility). The black and colored lines are similar in the sense that the overall peak shapes are the same amongst them. In other words, the black and colored lines are similar in their peaks but they differ in size (magnitude). These differ the most between regions 120 - 160. Between this region we see a large difference in peak heights which suggests that the colored structures are much more flexible and hints at their conformation being open or intermediate. (Intermediate stages in binding is when a structure is in its most flexible state.)